

Amanda Merkley

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Education

Carnegie Mellon University

Pittsburgh, PA

PHD ELECTRICAL ENGINEERING

August 2021 - present

- Advisor: Dr. Pulkit Grover

Colorado State University

Fort Collins, CO

BS COMPUTER ENGINEERING, BS MATHEMATICS

May 2020

- GPA: 3.97 (*magna cum laude*)

Research Experience

Carnegie Mellon University - Dept of Electrical & Computer Engineering

Pittsburgh, PA

ADVISOR: DR. PULKIT GROVER

August 2021 - Present

- *Structured generative model for latent representations*. Designed and implemented a novel variational autoencoder model for learning interpretable representations through concept-aligned and residual structure in neural data.
- *Foundation model representations for hunger characterization*. Pretrained a foundation model on 100GB of intracranial neural data and used its representations to characterize hunger-related neural activity.
- *Latent communication & causation in neural circuits*. Develop statistical model to identify latent communication and causal structure in neural systems.
- *Statistical analysis of neural data*. Identify effects of Deep Brain Stimulation on high-dimensional neural activity.
- *Cognitive experiment design*. Design and run experiments on 30+ human epilepsy patients implanted with depth electrodes.

Sandia National Laboratories - Cognitive & Emerging Computing

Albuquerque, NM

ADVISOR: DR. WILLIAM CHAPMAN

June 2025-August 2025

- *Redundant & synergistic statistical information flow*. Extended information-theoretic redundancy-synergy index to directed information to compute high-order information flow between neural populations engaged in navigation.

Massachusetts Institute of Technology - Center for Brains, Minds, & Machines

Boston, MA

ADVISOR: DR. NANCY KANWISHER

June 2019-August 2019

- *Object permanence in humans*. Designed and ran experiments in object permanence, analyzed MEG signals of 5 subjects.

Publications

A. Merkley and P. Grover. "Concept-localized generative representations." *Under review*, 2026.

Y. Guo, **A. Merkley**, S. Jaffee, A. C. Whiting, and P. Grover. "Identifying neural biomarkers of risk-taking from intracranial EEG recordings." *IEEE EMBS Neural Engineering*, 2025.

C. Goswami and **A. Merkley**. "Analytically deriving partial information decomposition for affine systems of stable and convolution-closed distributions." *Advances in Neural Information Processing Systems*, 2024.

A. Merkley and P. Grover. "Understanding neural population communication with latent channels." *Allerton Conference on Communication, Control, and Computing*, 2024.

A. Merkley, A. Y. Nam, Y. K. Hong, and P. Grover. "Message-relevant dimension reduction of neural populations." *IEEE International Symposium on Information Theory*, 2024.

Industry Experience

Western Digital Corporation

ASIC DEVELOPMENT ENGINEER

Longmont, CO

June 2020 - June 2021

- Verification of servo processor in enterprise hard drive ASIC in SystemVerilog and UVM.

Presentations

October 2024. *Characterizing high-order interactions during conflict processing in patients with epilepsy*. Poster: Society for Neuroscience 2024, Chicago, IL.

September 2024. *Understanding neural population communication with latent channels*. Talk: Allerton 2024, Urbana-Champaign, IL.

July 2024. *Message-relevant dimension reduction of neural populations*. Talk: ISIT 2024, Athens, Greece.

June 2023. *Computing unique information for Poisson sources*. Poster: Decomposing Multivariate Info in Complex Systems, Dresden, Germany.

Awards & Fellowships

2025 **URA-Sandia Grad Student Summer Fellowship**, Oak Ridge Inst. for Science & Education
Waibel Presidential Fellowship, Carnegie Mellon University, School of Computer Science

2023 **Exemplary Qualifying Exam Performance**, Carnegie Mellon University, ECE Department

2022 **NSF Graduate Research Fellowship Program**, National Science Foundation
Fritsch Family Fellowship, Carnegie Mellon University, ECE Department

2021 **NIH T-32 Neural Interfacing Traineeship**, National Institutes of Health

2019 **Astronaut Scholarship**, Astronaut Scholars Foundation
Goldwater Scholarship, Goldwater Foundation
Richard E. Merwin Student Scholarship, IEEE Computer Society

Service & Teaching

Summer 2024 **Summer Internship Mentor**, Mentored 2 undergraduate students

July 2024 **ISIT 2024 Neuroscience + Information Theory Workshop**, Workshop co-organizer

Spring 2024 **Teaching Assistant**, Information Theory Measures (18-753)

2021-2023 **ECE Graduate Organization**, Carnegie Mellon University, Treasurer

2017-2019 **IEEE Student Chapter**, Colorado State University, President